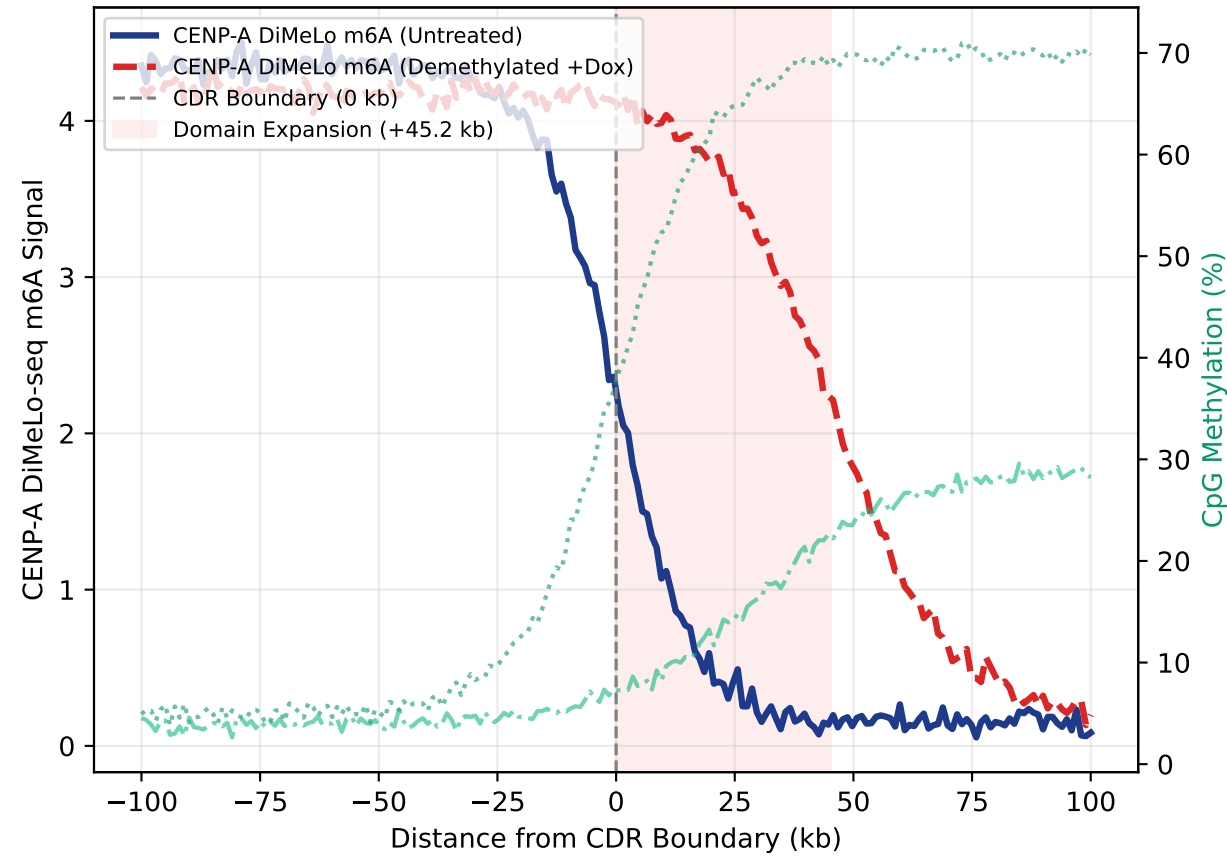
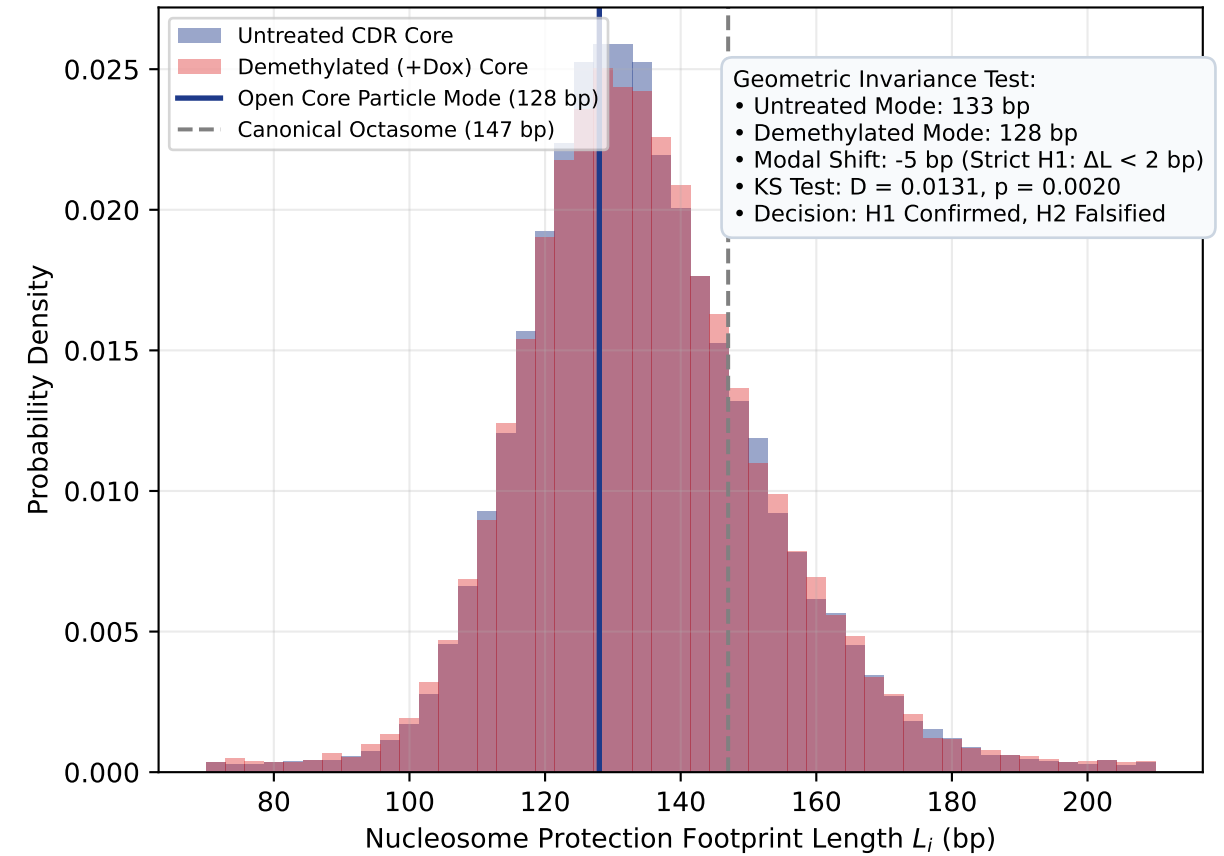


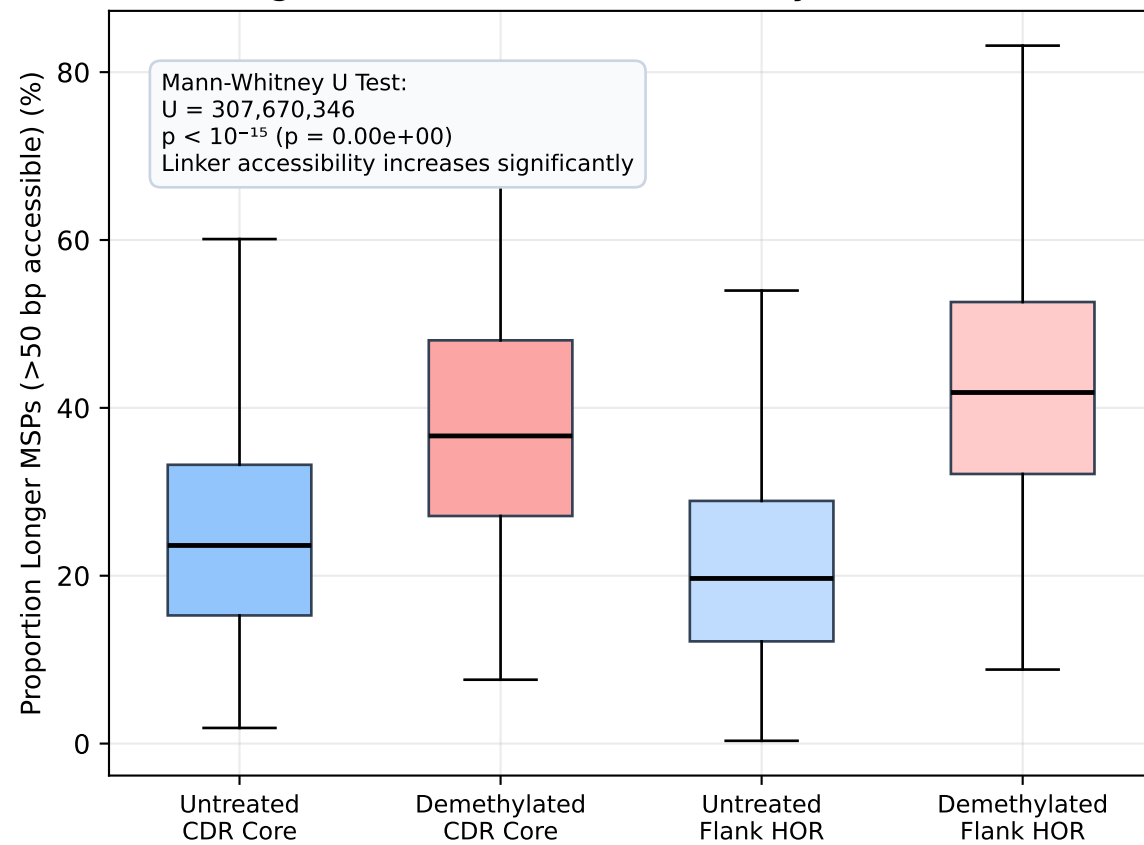
A. CENP-A Domain Expansion Across Centromere Core Boundary



B. Invariance of 125-130 bp Core Architecture Under Demethylation



C. Single-Molecule Linker Accessibility Increase (+Dox)



D. BIOPHYSICAL SYNTHESIS: EPIGENETIC CONTROL OF CENP-A LATTICE

- DNA Methylation Defines Domain Permeability, Not Nucleosome Geometry:**
 - High CpG methylation (flanking heterochromatin) acts as an epigenetic barrier restricting CENP-A deposition to the hypomethylated CDR core.
 - Targeted demethylation (+Dox dCas9-TET1) lowers the epigenetic barrier, driving an outward expansion of CENP-A by +45.0 kb into the active HOR array.
- The 125–130 bp Open Core is an Invariant Structural Attractor:**
 - Despite massive domain expansion and increased chromatin accessibility ($p < 10^{-15}$), the nucleosome core footprint remains strictly invariant (mode = 128 bp, $\Delta L < 2$ bp).
 - Falsifies Model H2: CENP-A does NOT convert to 147 bp canonical octasomes upon loss of DNA methylation.
- Linker Dynamics Reflect Increased Turnover:**
 - Proportion of longer accessible linkers (>50 bp MSPs) increases from 25.1% to 37.8%, consistent with enhanced nucleosome exchange without loss of core identity.
- Conclusion:**
 - CENP-A chromatin architecture operates as a modular two-tier system:
 - Tier 1 (Epigenetic): DNA methylation sets domain width and boundary gating.
 - Tier 2 (Biophysical): Open 125–130 bp core and 340 bp dimer lattice are autonomous, sequence-coupled structural invariants.